From question to research evidence

Ten real-paper research scenarios, observed through the normal application interface.

ArchimedesAI

Research walkthrough & reliability review

10/10 reports completed | 10/10 expected titles retrieved in the final observed runs

This report documents real live-provider runs, the defects they exposed, the improvements made, and the boundaries of the evidence. It is an engineering review, not an independent security audit or scientific validation certificate.

What was actually tested

Research questions were typed into the normal browser form. The app called live scholarly services and the configured Groq model. Screenshots capture the actual interface, not fabricated results.

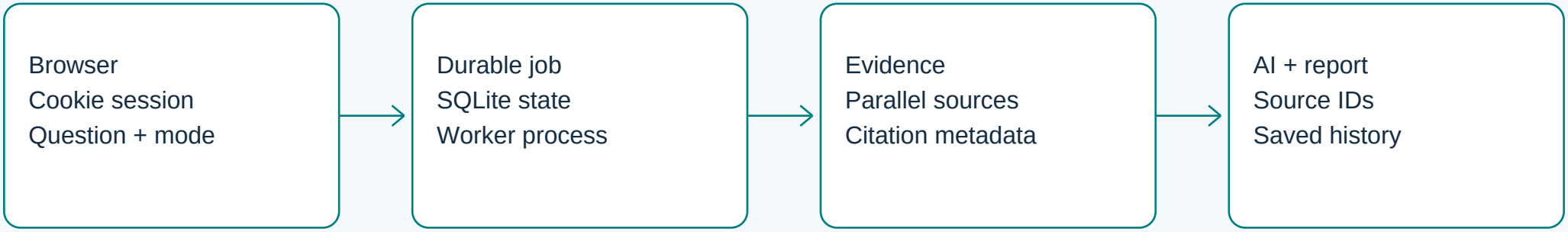
Ten target titles and review concerns were labelled in cases.json. The read-only collector joins each question to its latest saved report. Earlier failed or superseded attempts remain recorded in results.json; this is not a first-attempt success rate.

An isolated local study account/database kept the walkthrough separate from existing user research. Credentials, session cookies, encrypted keys and full published paper files are excluded from the report and repository.

Automated tests use synthetic providers and a separate database. Their results are reported separately from live retrieval and browser observations.

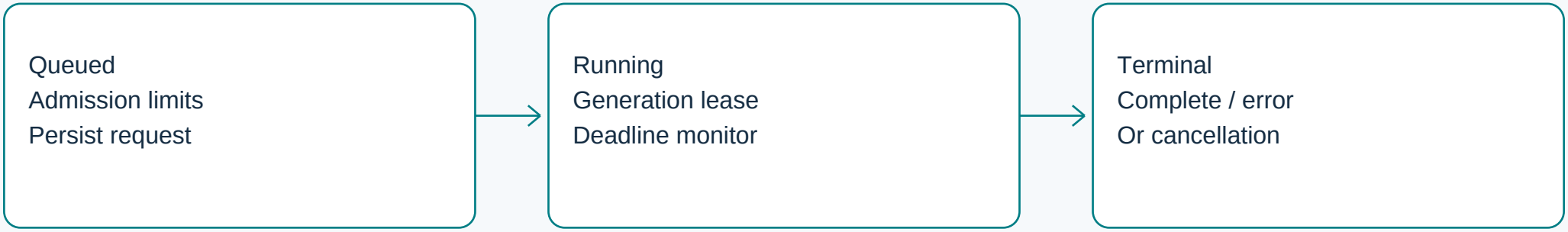
Claim review is a limited agent-assisted comparison of main ideas with linked original-paper descriptions. It does not measure the accuracy of every sentence, extraction cell, citation or generated manuscript.

How a research request moves



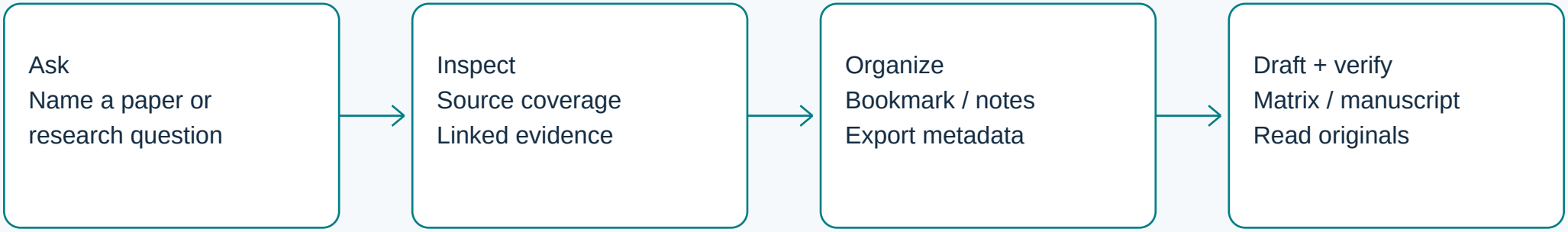
Progress is streamed from persisted events. A browser disconnect leaves the job running. A stale worker lease can trigger a bounded restart; completion and saved-report insertion happen atomically.

Recovery has explicit boundaries



Expired leases are reclaimable up to three attempts within the original deadline. Recovery repeats computation; it does not resume a provider socket. Cancellation terminates the owned worker, but cannot retract requests already received by an external provider.

Use the result as a starting point



The paper-search checkbox skips AI synthesis. With it unchecked, the app produces an AI-assisted synthesis. A confidence badge or matched citation identity does not establish that a claim is scientifically correct.

Ten labelled research scenarios

UC01 Transformer attention | 16 papers | 16.6s | target found

UC02 BERT pretraining | 6 papers | 8.3s | target found

UC03 Retrieval-augmented generation | 14 papers | 13.9s | target found

UC04 Residual image recognition | 11 papers | 18.3s | target found

UC05 Parameter-efficient adaptation | 10 papers | 18.0s | target found

UC06 Graph neural networks | 21 papers | 22.7s | target found

UC07 Approximate vector search | 5 papers | 14.4s | target found

UC08 Protein structure prediction | 19 papers | 20.5s | target found

UC09 Programmable genome editing | 14 papers | 14.0s | target found

UC10 Quantum error correction | 14 papers | 22.9s | target found

Final-run server pipeline time: median 17.3s; range 8.3-22.9s. Browser rendering and delayed-proxy delivery can take longer. Ten selected questions do not establish general relevance or uptime.

Transformer attention

Observed question, expected evidence and result

QUESTION

Find "Attention Is All You Need" and explain self-attention in the Transformer architecture.

EXPECTED PAPER

Attention Is All You Need
https://arxiv.org/abs/1706.03762

OBSERVED

16 papers; 16.6 seconds; expected title found at rank 1. 0 retrieved citation links.

REVIEW FOCUS

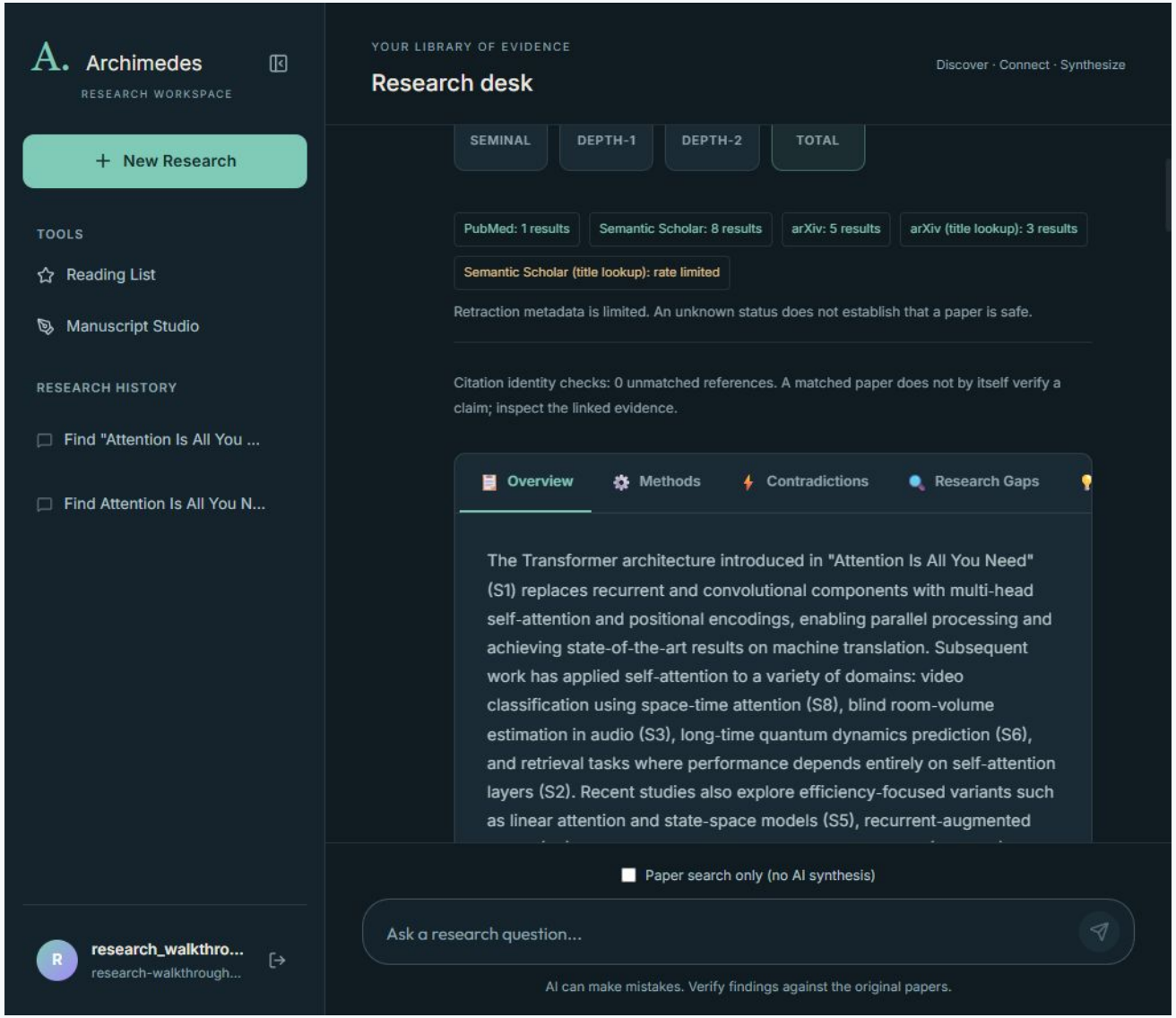
The original proposes an attention-based architecture without recurrent or convolutional sequence modelling; related-paper summaries must not be mistaken for the original.

SOURCE COVERAGE

PubMed: results (1); Semantic Scholar: results (8); arXiv: results (5); arXiv (title lookup): results (3); Semantic Scholar (title lookup): rate_limited (0)

Observed application output

Transformer attention - actual browser screenshot; original layout preserved



BERT pretraining

Observed question, expected evidence and result

QUESTION

Find "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" and explain its pretraining objectives.

EXPECTED PAPER

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
https://arxiv.org/abs/1810.04805

OBSERVED

6 papers; 8.3 seconds; expected title found at rank 1. 0 retrieved citation links.

REVIEW FOCUS

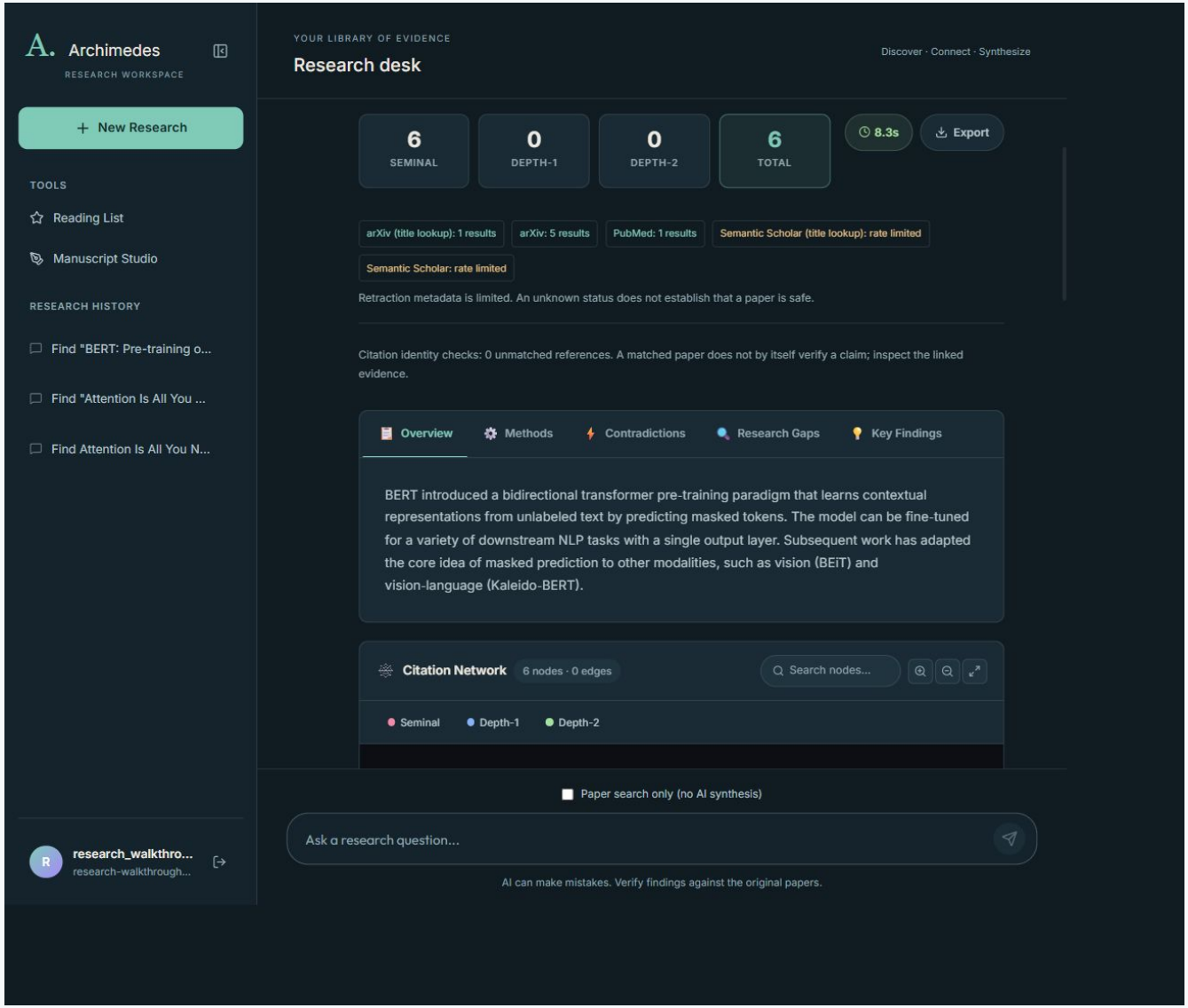
Distinguish bidirectional pretraining and masked language modelling from generic Transformer discussion.

SOURCE COVERAGE

arXiv (title lookup): results (1); arXiv: results (5); PubMed: results (1); Semantic Scholar (title lookup): rate_limited (0); Semantic Scholar: rate_limited (0)

Observed application output

BERT pretraining - actual browser screenshot; original layout preserved



Retrieval-augmented generation

Observed question, expected evidence and result

QUESTION

Find "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" and explain the retrieval and generation components.

EXPECTED PAPER

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
https://arxiv.org/abs/2005.11401

OBSERVED

14 papers; 13.9 seconds; expected title found at rank 1. 8 retrieved citation links.

REVIEW FOCUS

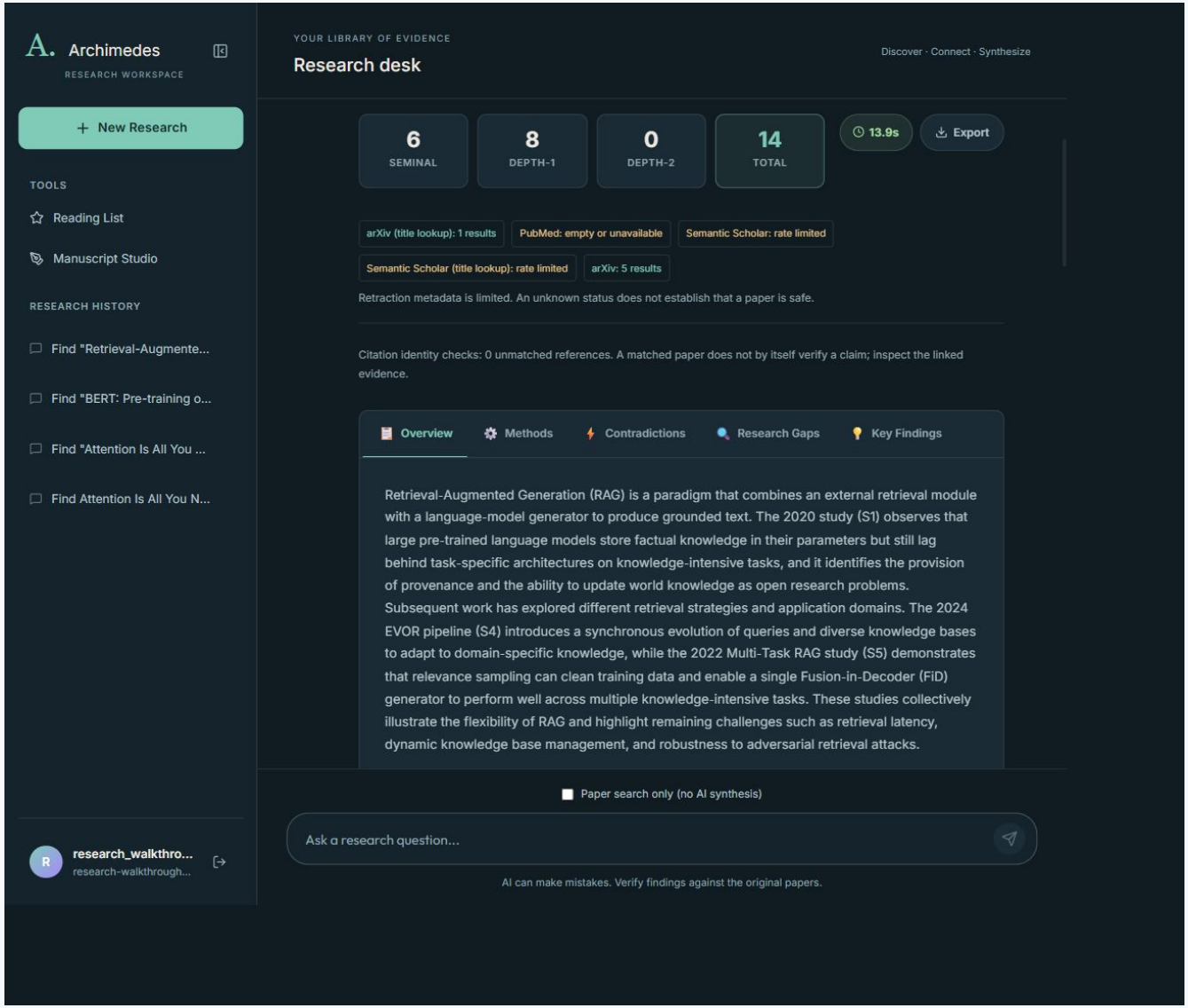
Separate retrieved non-parametric evidence from the generator; avoid inventing benchmark improvements.

SOURCE COVERAGE

arXiv (title lookup): results (1); PubMed: empty_or_unavailable (0); Semantic Scholar: rate_limited (0); Semantic Scholar (title lookup): rate_limited (0); arXiv: results (5)

Observed application output

Retrieval-augmented generation - actual browser screenshot; original layout preserved



Residual image recognition

Observed question, expected evidence and result

QUESTION

Find "Deep Residual Learning for Image Recognition" and explain residual connections.

EXPECTED PAPER

Deep Residual Learning for Image Recognition
https://arxiv.org/abs/1512.03385

OBSERVED

11 papers; 18.3 seconds; expected title found at rank 1. 0 retrieved citation links.

REVIEW FOCUS

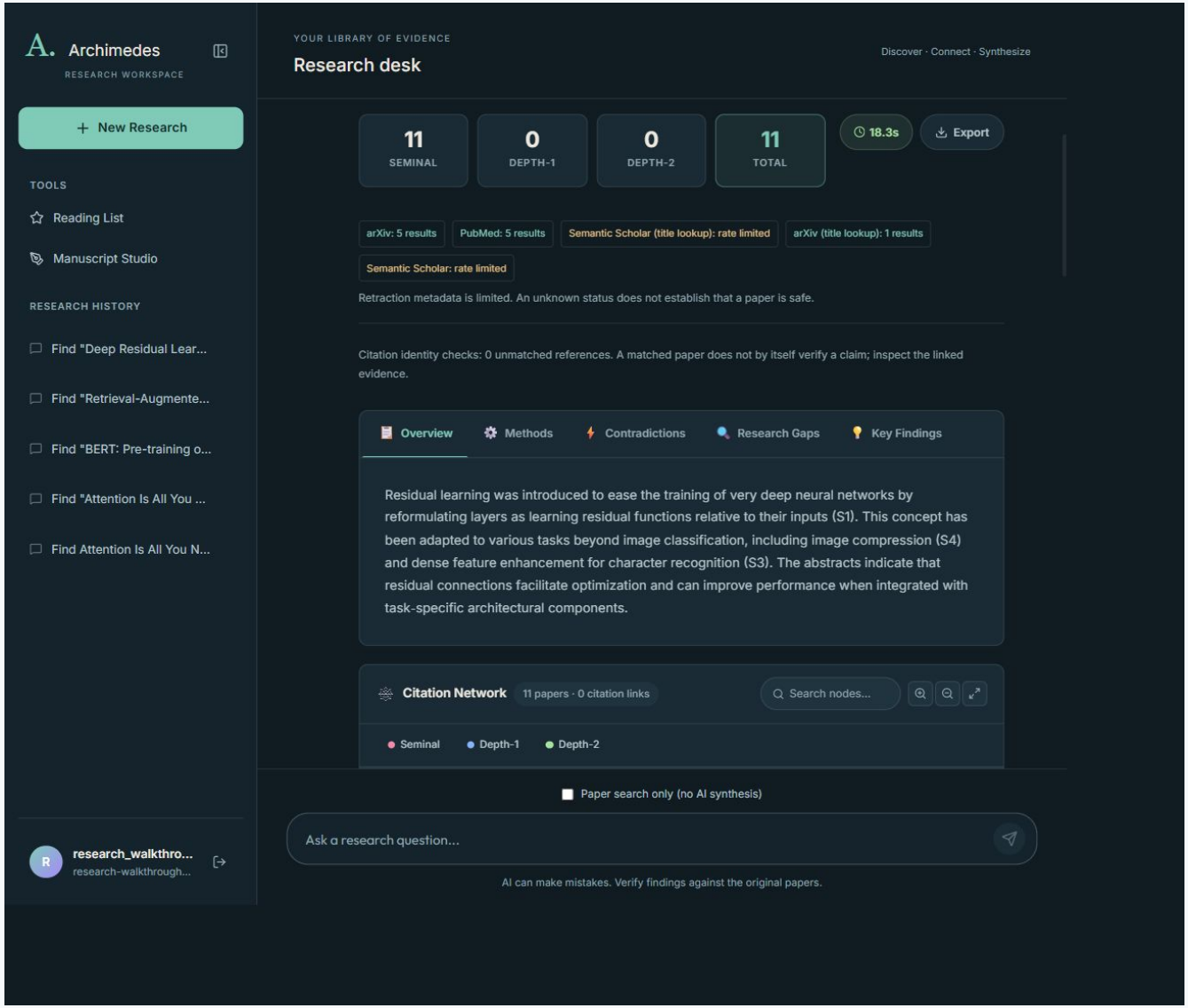
Explain residual learning and optimization without claiming every deeper network is automatically better.

SOURCE COVERAGE

arXiv: results (5); PubMed: results (5); Semantic Scholar (title lookup): rate_limited (0); arXiv (title lookup): results (1); Semantic Scholar: rate_limited (0)

Observed application output

Residual image recognition - actual browser screenshot; original layout preserved



Parameter-efficient adaptation

Observed question, expected evidence and result

QUESTION

Find "LoRA: Low-Rank Adaptation of Large Language Models" and explain which parameters are trained.

EXPECTED PAPER

LoRA: Low-Rank Adaptation of Large Language Models
https://arxiv.org/abs/2106.09685

OBSERVED

10 papers; 18.0 seconds; expected title found at rank 1. 0 retrieved citation links.

REVIEW FOCUS

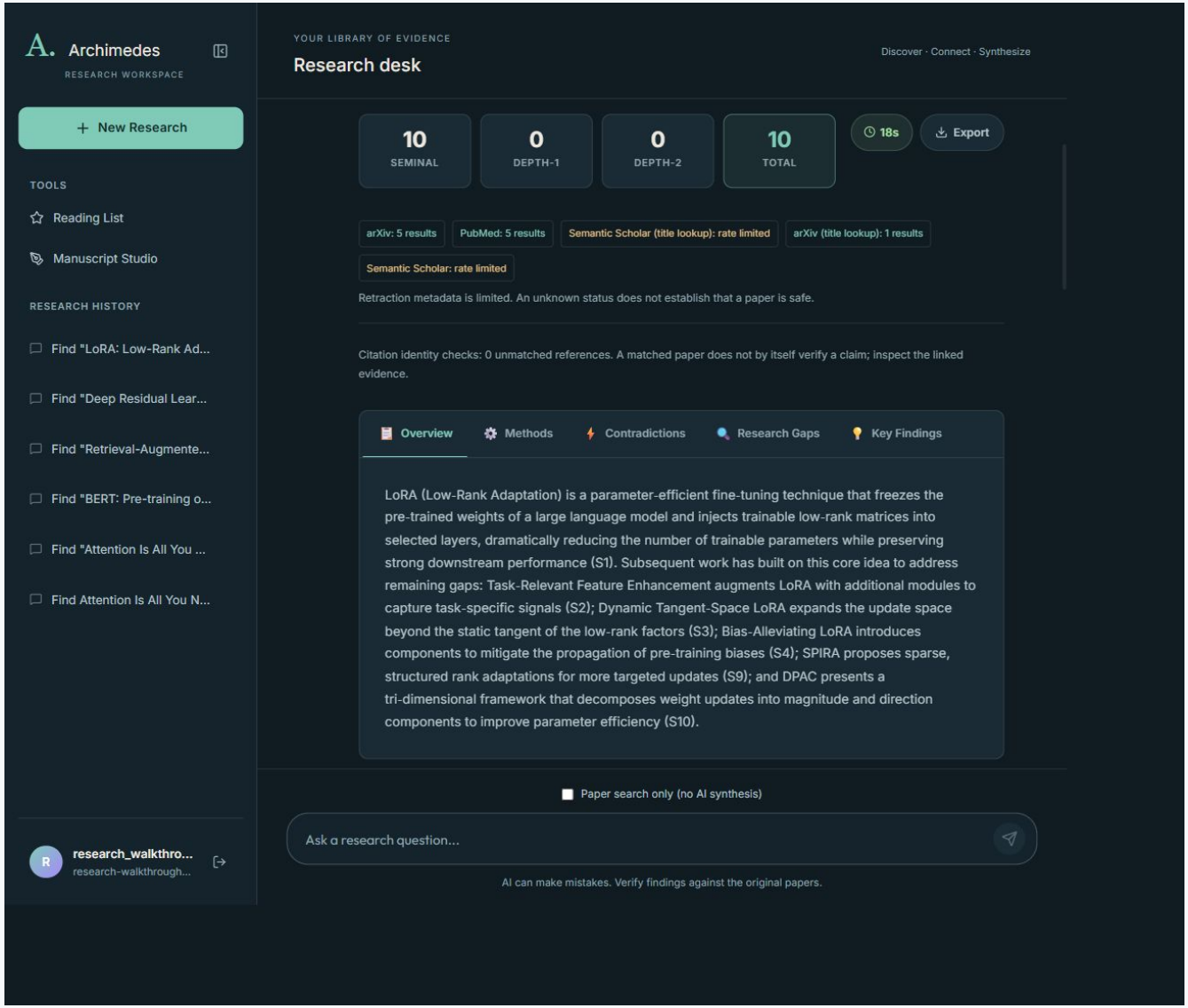
Distinguish frozen pretrained weights from trainable low-rank update matrices; verify any numeric savings.

SOURCE COVERAGE

arXiv: results (5); PubMed: results (5); Semantic Scholar (title lookup): rate_limited (0); arXiv (title lookup): results (1); Semantic Scholar: rate_limited (0)

Observed application output

Parameter-efficient adaptation - actual browser screenshot; original layout preserved



Graph neural networks

Observed question, expected evidence and result

QUESTION

Find "Graph Attention Networks" and explain neighbourhood attention.

EXPECTED PAPER

Graph Attention Networks
https://arxiv.org/abs/1710.10903

OBSERVED

21 papers; 22.7 seconds; expected title found at rank 1. 8 retrieved citation links.

REVIEW FOCUS

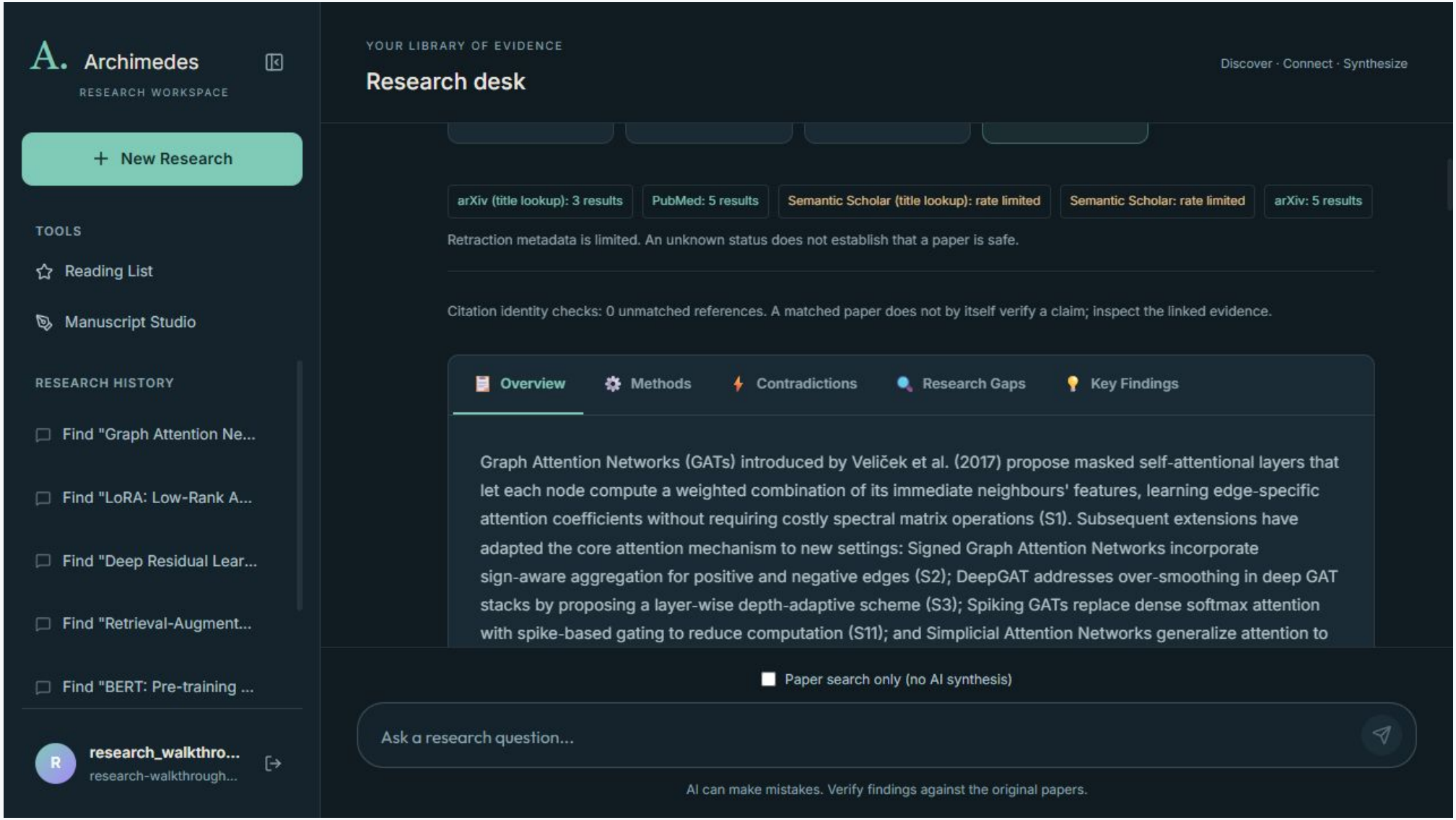
Neighbour weighting through attention should be grounded in the retrieved GAT paper.

SOURCE COVERAGE

arXiv (title lookup): results (3); PubMed: results (5); Semantic Scholar (title lookup): rate_limited (0); Semantic Scholar: rate_limited (0); arXiv: results (5)

Observed application output

Graph neural networks - actual browser screenshot; original layout preserved



Approximate vector search

Observed question, expected evidence and result

QUESTION

Find "Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs" and explain the graph hierarchy.

EXPECTED PAPER

Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs
https://arxiv.org/abs/1603.09320

OBSERVED

5 papers; 14.4 seconds; expected title found at rank 1. 0 retrieved citation links.

REVIEW FOCUS

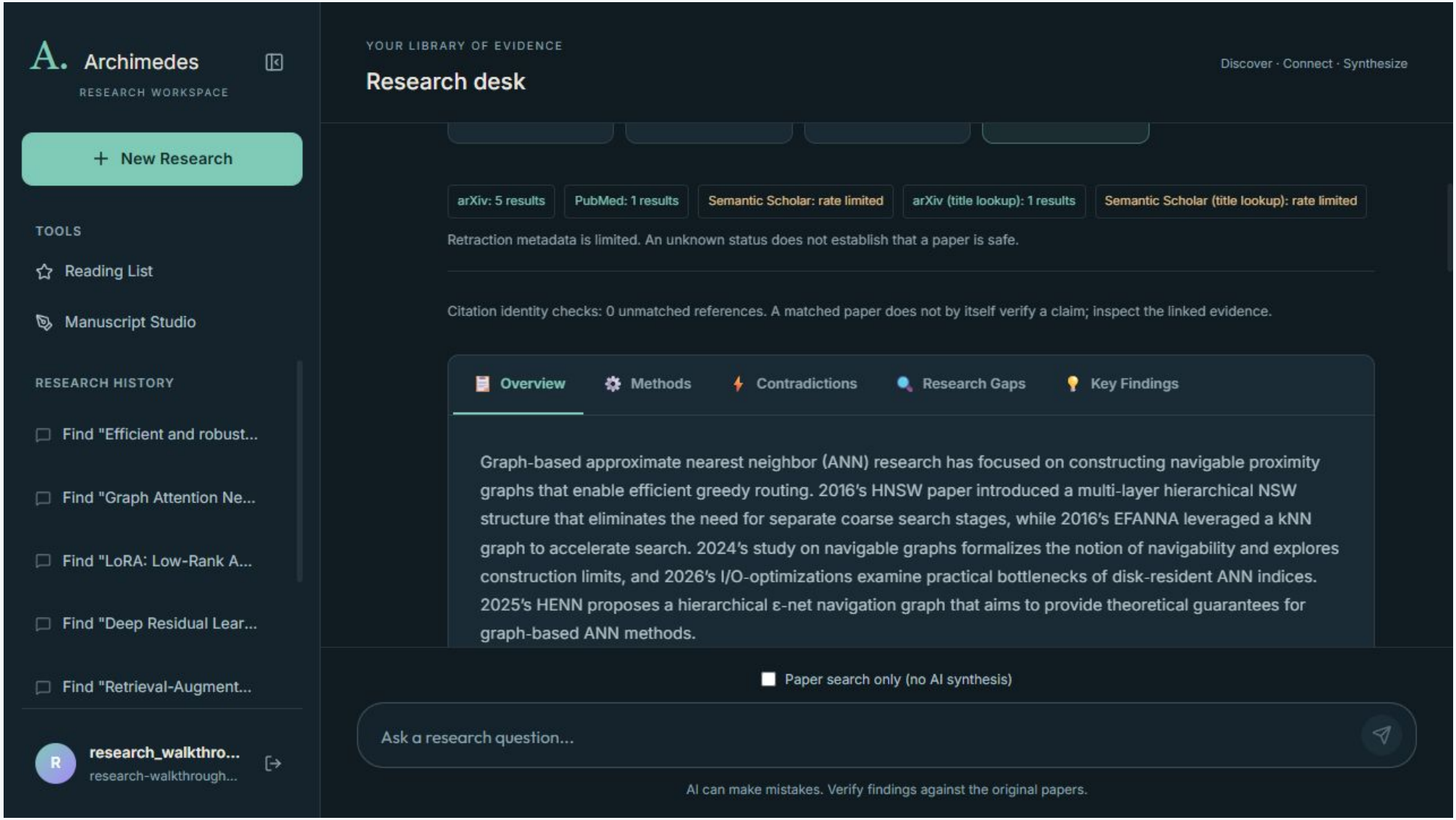
Explain approximate search and hierarchical navigable graphs; do not imply exact nearest-neighbour guarantees.

SOURCE COVERAGE

arXiv: results (5); PubMed: results (1); Semantic Scholar: rate_limited (0); arXiv (title lookup): results (1); Semantic Scholar (title lookup): rate_limited (0)

Observed application output

Approximate vector search - actual browser screenshot; original layout preserved



Protein structure prediction

Observed question, expected evidence and result

QUESTION

Find "Highly accurate protein structure prediction with AlphaFold" and explain the structure prediction approach.

EXPECTED PAPER

Highly accurate protein structure prediction with AlphaFold
https://doi.org/10.1038/s41586-021-03819-2

OBSERVED

19 papers; 20.5 seconds; expected title found at rank 1. 8 retrieved citation links.

REVIEW FOCUS

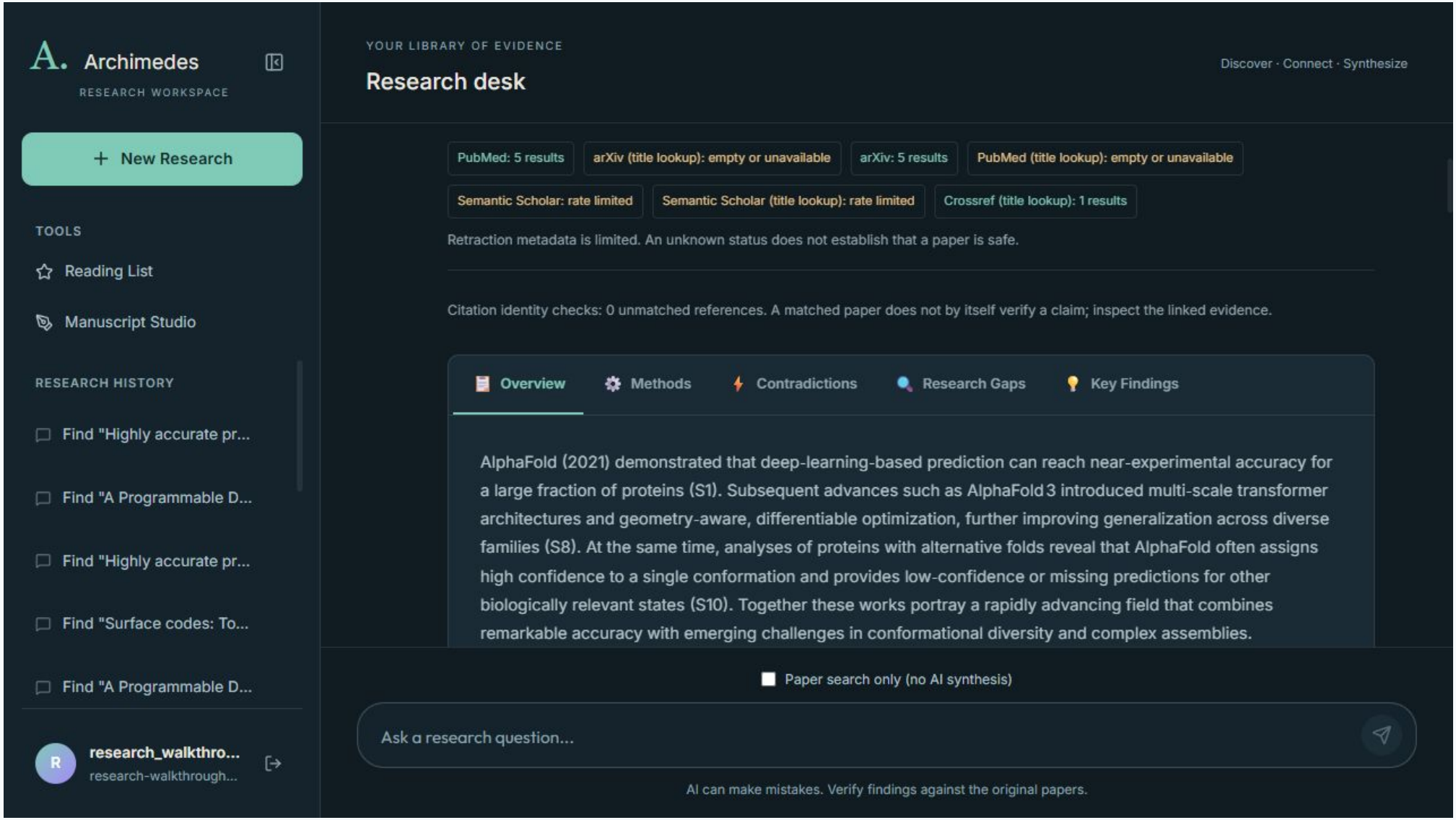
Describe a computational structure-prediction result; do not turn it into clinical or experimental validation claims.

SOURCE COVERAGE

PubMed: results (5); arXiv (title lookup): empty_or_unavailable (0); arXiv: results (5); PubMed (title lookup): empty_or_unavailable (0); Semantic Scholar: rate_limited (0); Semantic Scholar (title lookup): rate_limited (0); Crossref (title lookup): results (1)

Observed application output

Protein structure prediction - actual browser screenshot; original layout preserved



Programmable genome editing

Observed question, expected evidence and result

QUESTION

Find "A Programmable Dual-RNA-Guided DNA Endonuclease in Adaptive Bacterial Immunity" and explain the role of guide RNA.

EXPECTED PAPER

A Programmable Dual-RNA-Guided DNA Endonuclease in Adaptive Bacterial Immunity
https://doi.org/10.1126/science.1225829

OBSERVED

14 papers; 14.0 seconds; expected title found at rank 1. 8 retrieved citation links.

REVIEW FOCUS

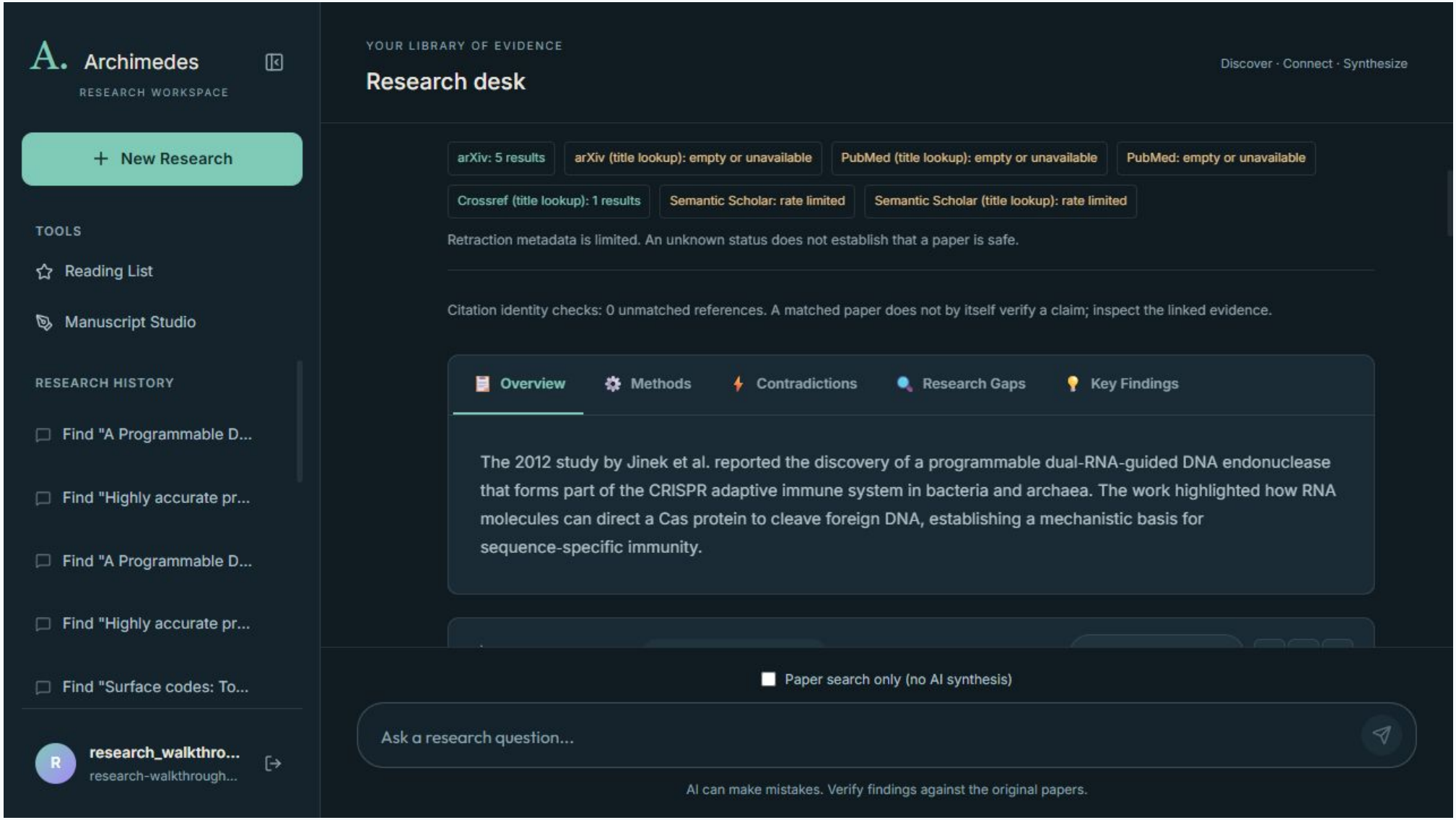
Check the paper identity and RNA-guided mechanism; no medical treatment or safety conclusions are evaluated.

SOURCE COVERAGE

arXiv: results (5); arXiv (title lookup): empty_or_unavailable (0); PubMed (title lookup): empty_or_unavailable (0); PubMed: empty_or_unavailable (0); Crossref (title lookup): results (1); Semantic Scholar: rate_limited (0); Semantic Scholar (title lookup): rate_limited (0)

Observed application output

Programmable genome editing - actual browser screenshot; original layout preserved



Quantum error correction

Observed question, expected evidence and result

QUESTION

Find "Surface codes: Towards practical large-scale quantum computation" and explain the error-correction setting.

EXPECTED PAPER

Surface codes: Towards practical large-scale quantum computation
https://arxiv.org/abs/1208.0928

OBSERVED

14 papers; 22.9 seconds; expected title found at rank 1. 8 retrieved citation links.

REVIEW FOCUS

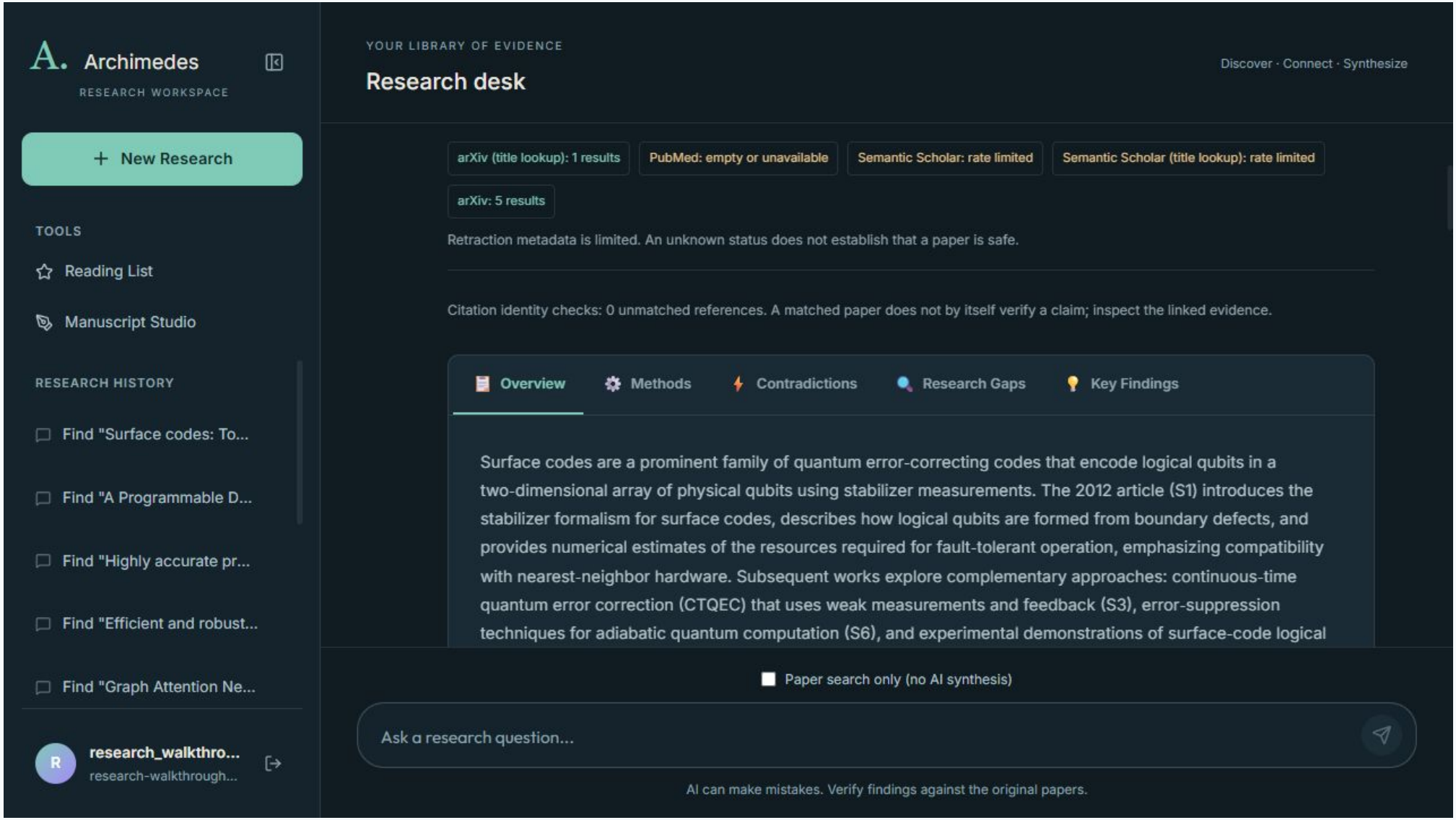
Distinguish theoretical error-correction requirements from a claim of deployed fault-tolerant quantum computing.

SOURCE COVERAGE

arXiv (title lookup): results (1); PubMed: empty_or_unavailable (0); Semantic Scholar: rate_limited (0); Semantic Scholar (title lookup): rate_limited (0); arXiv: results (5)

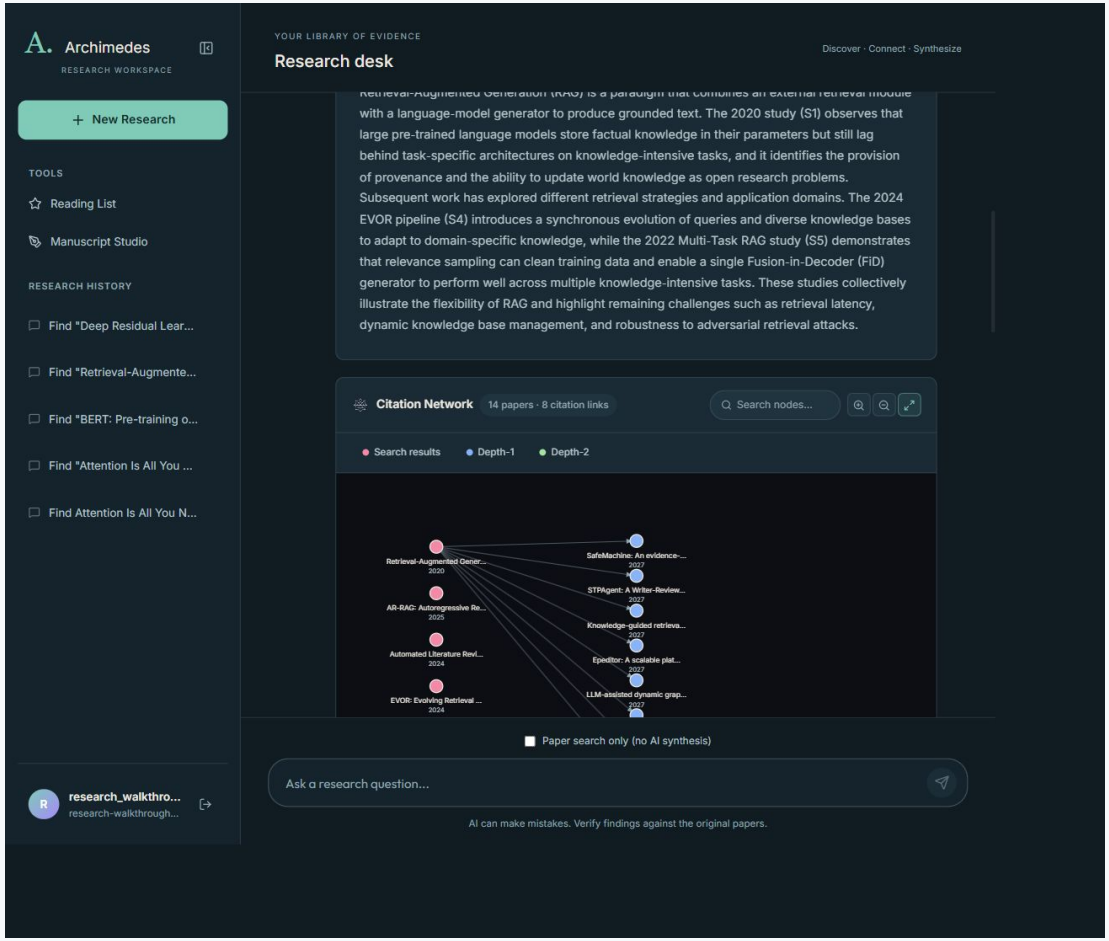
Observed application output

Quantum error correction - actual browser screenshot; original layout preserved



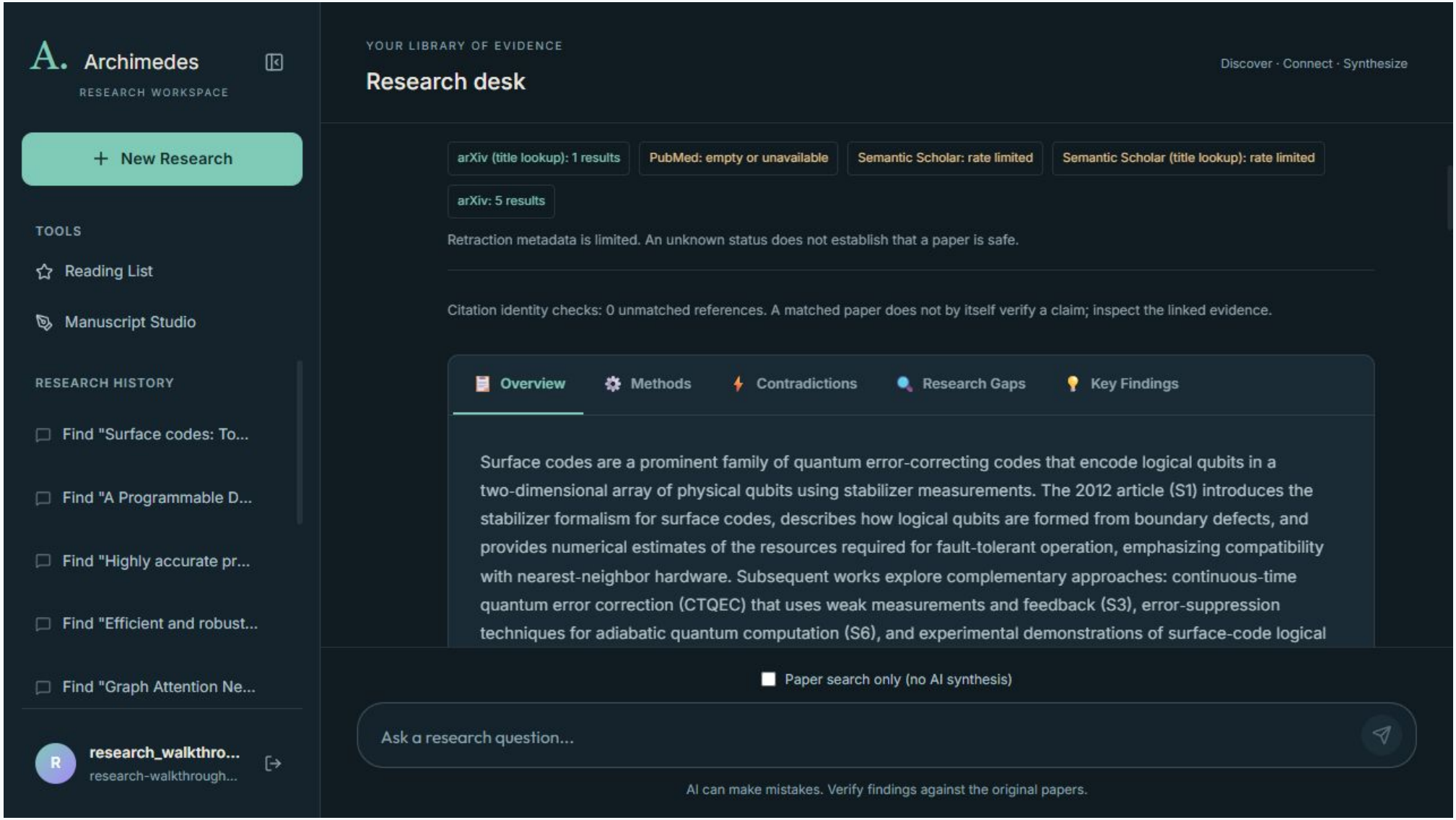
Papers can exist without retrieved links

The old canvas could render blank even when nodes existed. The replacement uses deterministic SVG positions, visible labels, keyboard-selectable papers and explicit no-link messaging. Only retrieved citation relationships become arrows. Zero links means unavailable or absent relationship data, not that the papers are unrelated.



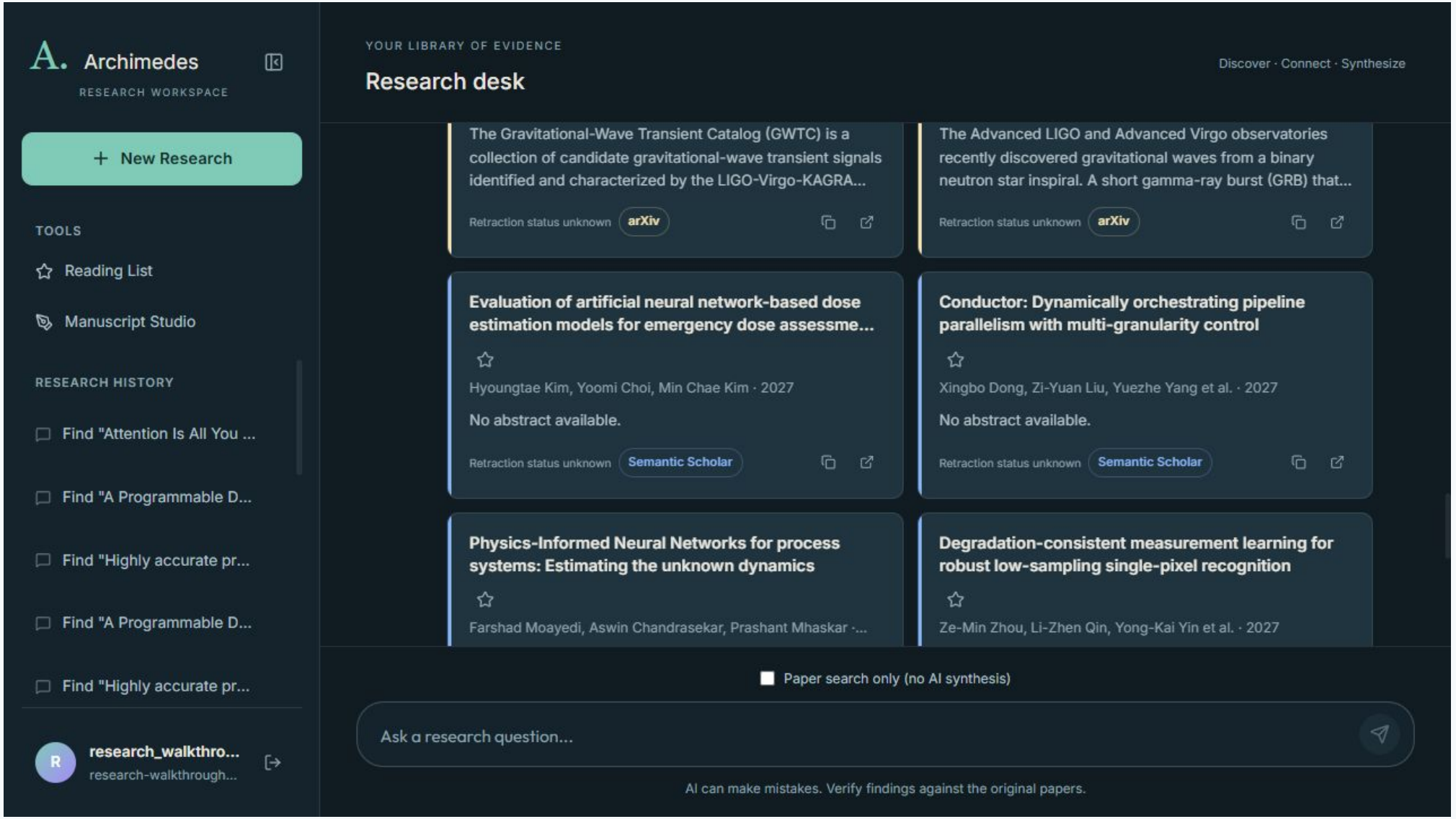
Saved history survives browser reload

Passed: the active surface-code research reconnected after a normal browser reload and displayed its saved report.



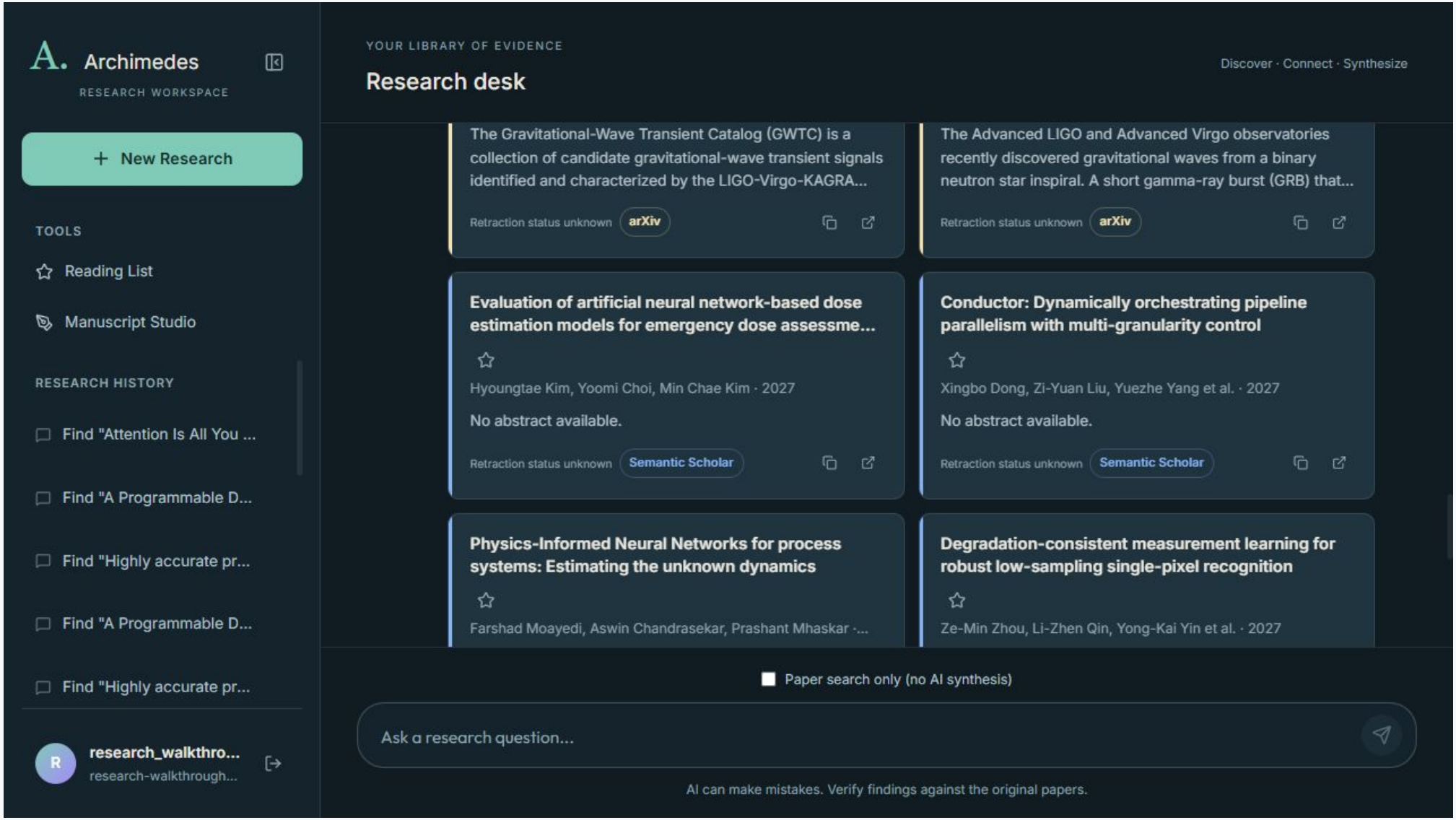
Buffering, split events and server restart

Passed with manual reconnection: a local proxy buffered two seconds and split SSE bytes into 37-byte chunks with 5 ms delays. A server restart interrupted the connection; the UI retained the job and reload recovered the metadata-only result.



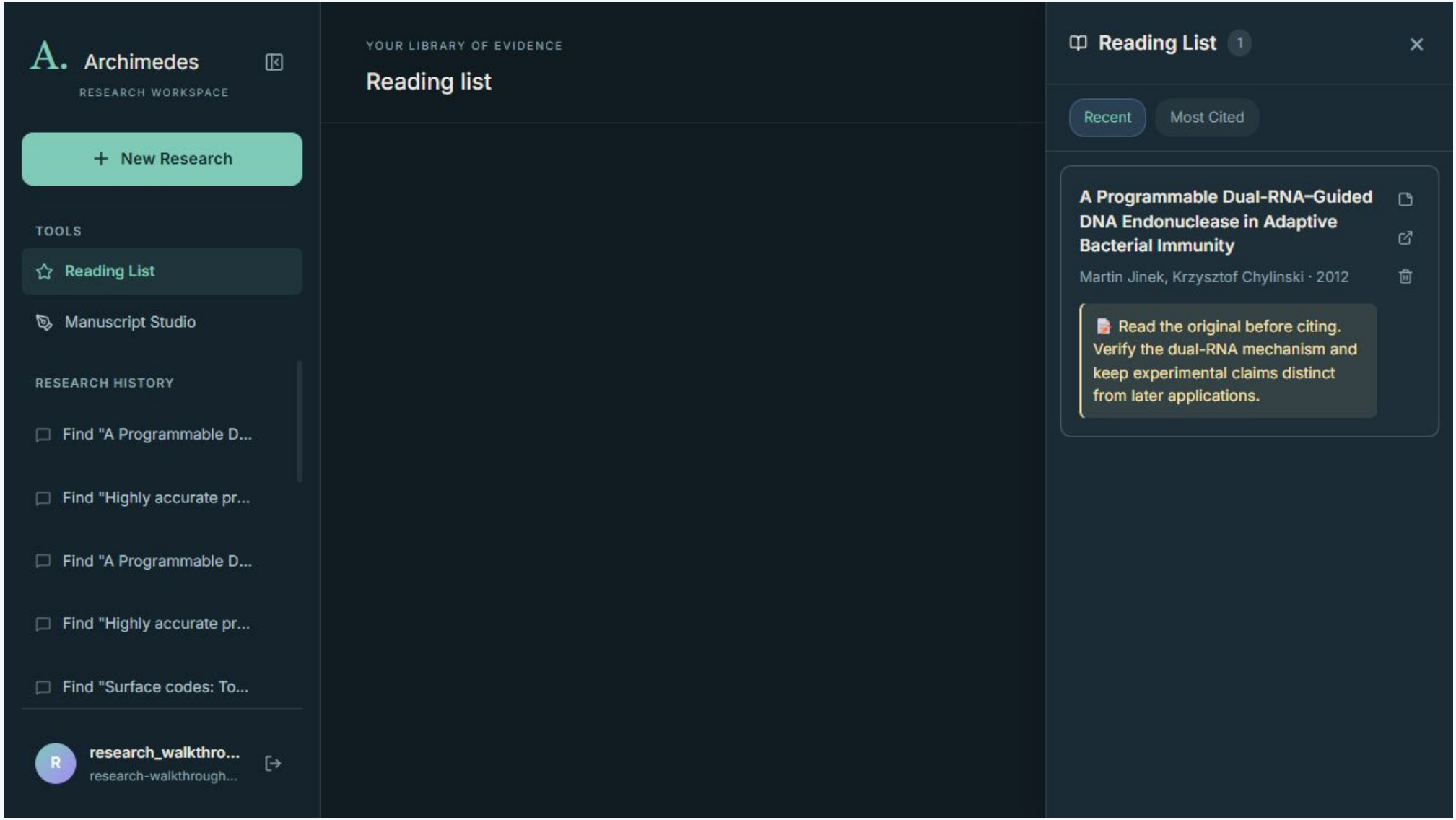
Paper-search mode

Passed: the recovered report explicitly says no AI synthesis was requested. The checkbox affects synthesis, not whether papers or citation metadata can be searched.



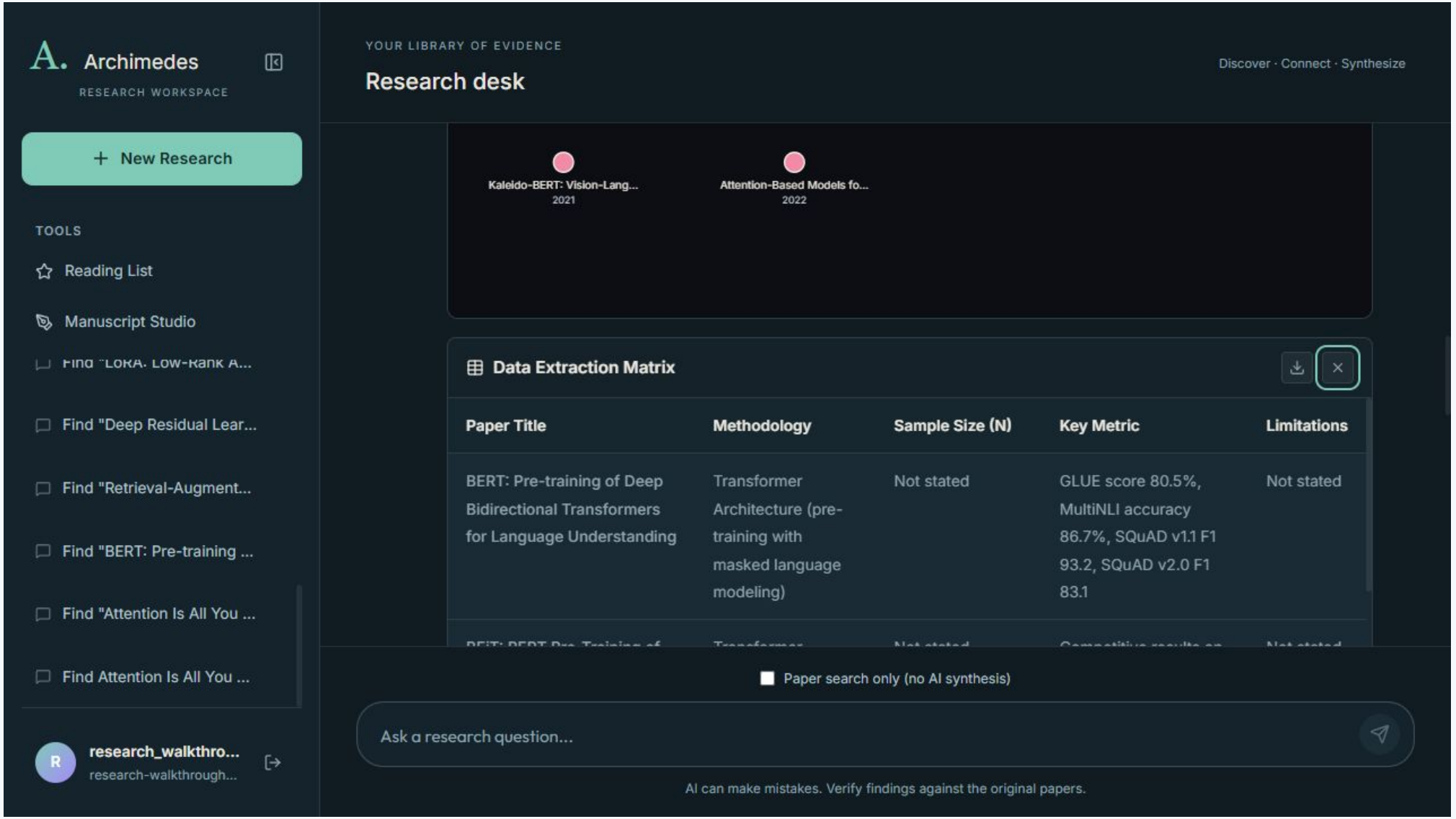
Reading list and research notes

Passed: the original CRISPR paper was bookmarked through its paper card and a verification note was saved in the reading list.



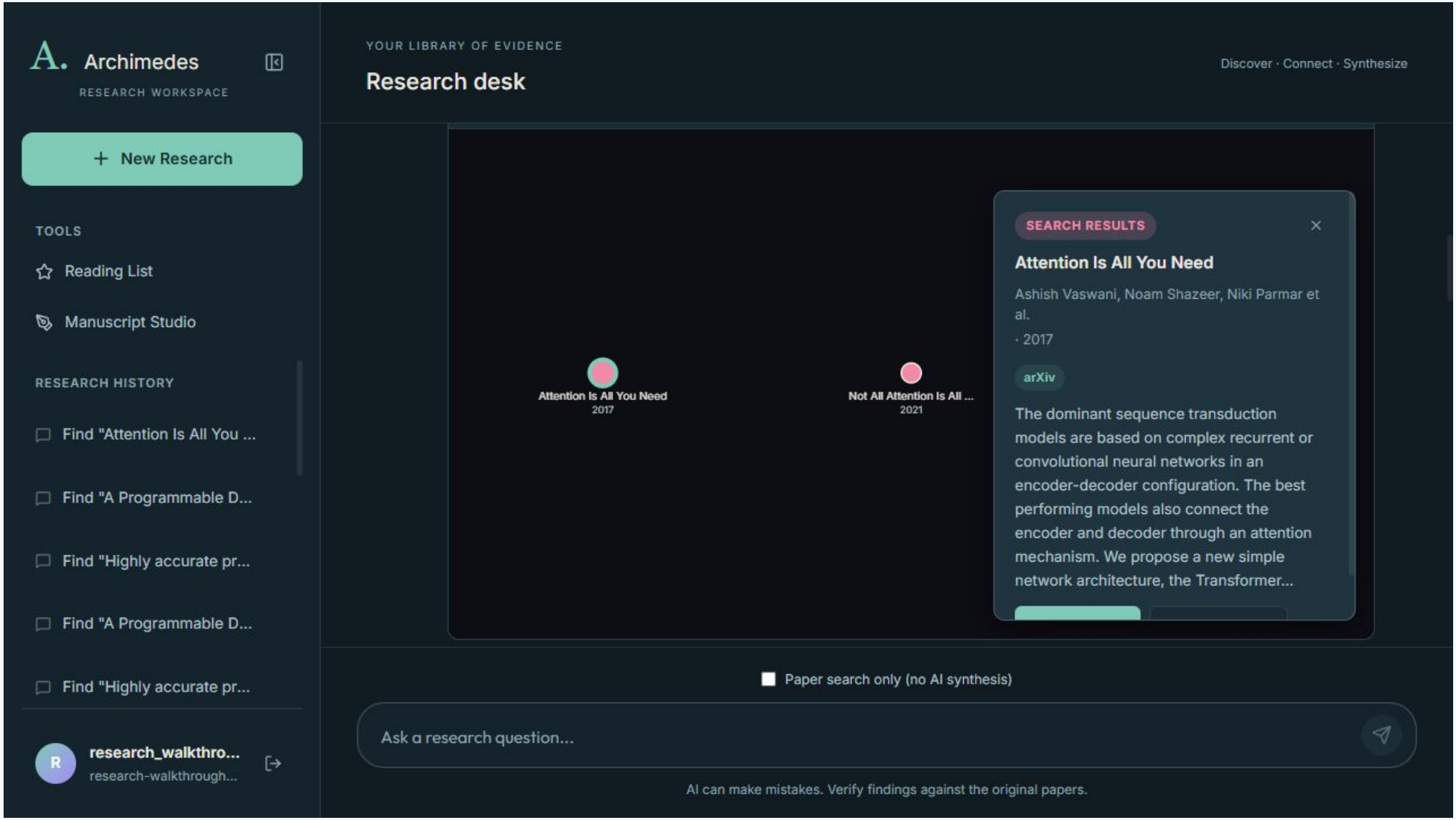
Evidence extraction matrix

Passed for output generation: six BERT-related rows appeared. The BERT benchmark values match its original abstract; missing fields were marked Not stated. Other rows were not exhaustively validated.



Keyboard-accessible paper graph

Passed: filtering for Attention Is All You Need and pressing Enter on its graph node opened the original paper details. Zero-link results remained visible.



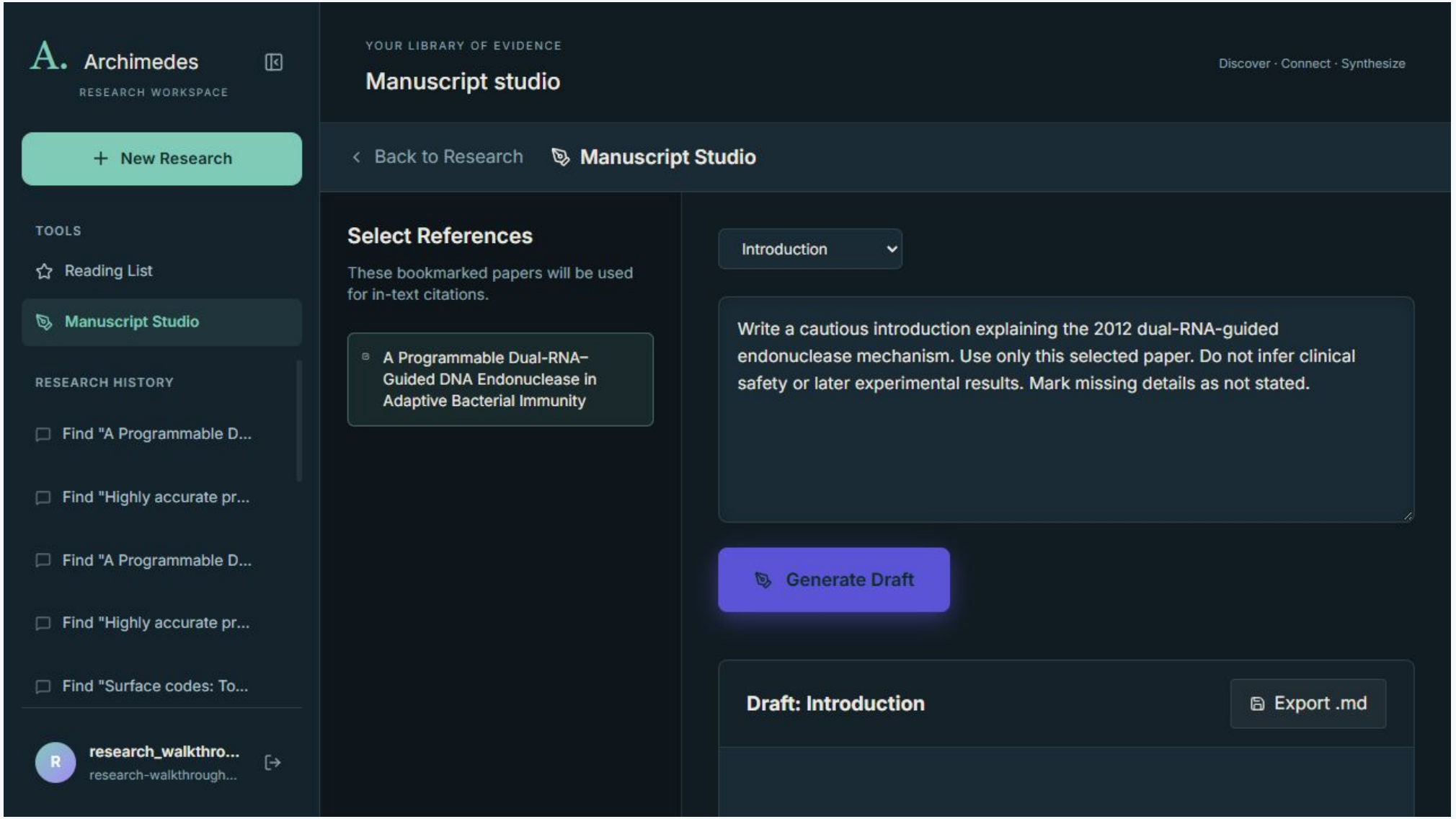
BibTeX export

Passed: the normal Export menu produced archimedes_3.bib. The downloaded file contained the BERT title, authors, year and paper URL.

The file was opened from the browser download location and its bibliographic fields inspected. Publication version/year conventions still need checking for a final bibliography. Other export formats have automated endpoint coverage; this study does not claim every format was manually downloaded.

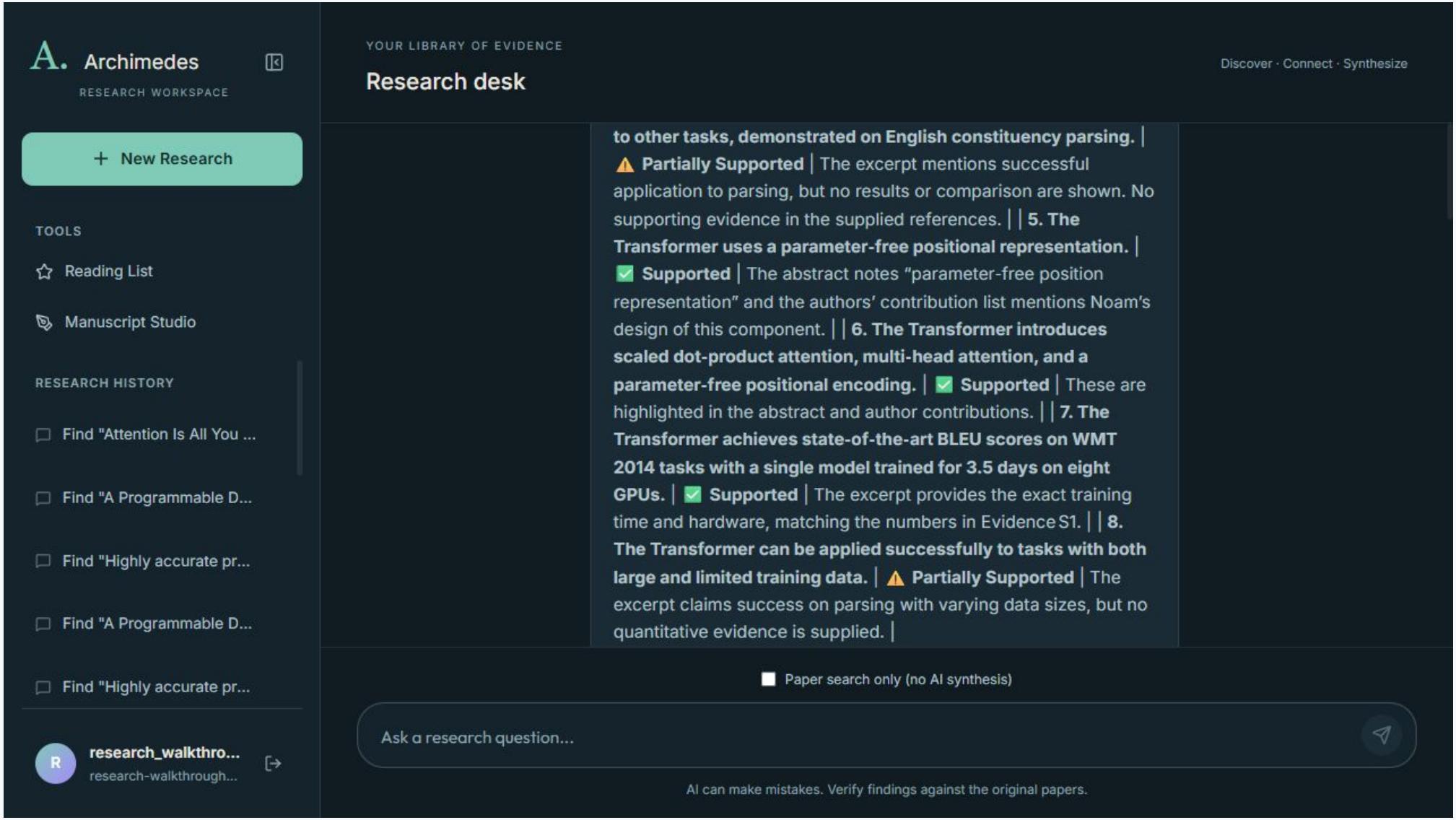
Manuscript drafting and its limits

Output generated, scientific review required: a CRISPR introduction appeared, but one sentence blurred PAM recognition and RNA roles, and missing excerpt details were phrased as absent study work. Instructions were tightened after this observation.



Original-paper PDF upload

Passed for file parsing and output generation: a 15-page Transformer paper was uploaded through the normal file chooser. The revised output distinguishes excerpt limitations and asks questions about the full paper; scientific accuracy remains unvalidated.



Useful outputs still need source review

Selected main ideas in the eight arXiv cases agree with the original abstract descriptions: attention-based Transformers, bidirectional BERT, retrieval plus generation, residual learning, frozen LoRA weights, neighbourhood attention, hierarchical approximate search and stabilizer-based surface codes.

Completeness is weaker than topic relevance. The BERT overview omits next-sentence prediction. The RAG overview does not fully detail the dense Wikipedia index or sequence/token variants. The CRISPR overview does not fully explain the two RNA components. Ten title hits do not establish ten complete scientific answers.

The CRISPR manuscript blurred RNA roles and PAM recognition and inferred absent study work from short excerpts. The first uploaded-PDF comparison recommended later papers as missing references and treated excerpt omissions as paper omissions. These failures are recorded rather than counted as scientific passes.

Drafting and PDF-comparison prompts now prohibit those unsupported inferences, and the interface labels the outputs provisional. This does not guarantee correctness. A larger labelled dataset and independent domain-expert review remain necessary.

AlphaFold and CRISPR identity was checked with retrieved metadata; publisher full text was not independently inspected through the web tool. See docs/user-study/CLAIM_REVIEW.md for case-by-case checks and the exact scope of source access.

What the reliability checks establish

Backend: 53 tests. Frontend: 7 tests. Lint, type checking and production build are checked separately. Automated provider responses are synthetic and do not contribute to live-paper success metrics.

Persistence: 25,000 saved reports backed up and restored with integrity/hash verification. Eight concurrent writers preserve 400 bookmark records. Additive migrations are rolled back and reapplied without losing existing research.

Failure handling: simulated SQLITE_FULL rolls back safely; database errors return sanitized 503 responses; corrupt backups are rejected. This does not reproduce a physically full disk or every filesystem failure.

Input/security: a seeded corpus of 100 malformed JSON bodies and 100 malformed PDFs produces client errors, plus adversarial usernames, private-path and security-header checks. This is bounded fuzzing, not an independent audit.

Jobs/models: persisted event replay, stale-lease recovery, cross-process rate windows, owned-process cancellation, retired-model filtering, output truncation fallback and title preservation have regression coverage.

Streaming/graph: split UTF-8 and JSON frames, malformed/truncated events, cancellation and finite node layouts are tested. The graph bundle decreased from approximately 194 KB to 8.47 KB after replacing the physics canvas.

Failures were part of the evaluation

Initial live synthesis failed because the previously configured model identifiers were unavailable. Model discovery now filters configured fallbacks against the provider catalogue; completion length and context are bounded.

A broad Transformer search initially returned related work rather than the original. Preserving explicitly named titles improved retrieval. Biomedical misses exposed limits in keyword and source coverage; direct PubMed and exact-title Crossref lookups were added, and reruns are disclosed.

Provider rate limits occurred during the study. Shared pacing and bounded retries reduce bursts but cannot guarantee access. Source coverage remains visible and metadata failures remain unknown.

An oversized chunked request initially returned 400 because parsing swallowed the size-limit exception. Pre-parser bounded spooling now produces 413. The blank citation canvas was independently fixed rather than explained away as missing links.

How reliable is it now?

Suitable for a local research-assistance demonstration: named-paper discovery, source inspection, saved history and interactive evidence exploration are backed by live browser observations and regression tests.

Not yet established: a production availability SLA, broad relevance benchmark, independent penetration audit, expert-labelled scientific accuracy, or exhaustive extraction and manuscript quality. A matched source ID is not proof of entailment.

Retraction status remains incomplete. Crossref and OpenAlex may share upstream metadata, and a missing notice is not evidence of safety. Metadata dates may differ between preprint and journal versions.

Operational limits include one-host SQLite coordination, no automatic job-event retention, bounded retry recovery and unverified real deployment/proxy/storage behavior. Configure retention, backups, monitoring and expert review before production.

Deployment, HTTPS configuration and public hosting were deliberately excluded at the user's request. The repository contains controls and test evidence, not a claim that every external assurance is complete.

Evidence files and verification commands

docs/user-study/cases.json: predefined questions, target titles and review concerns. results.json: latest saved outputs, metadata, provider observations and all job attempts. browser-observations.json: UI-captured synthesis and screenshots.

python scripts/collect_user_study.py reads the isolated study database; it does not initiate research. python scripts/build_walkthrough_report.py generates this PDF from collected evidence.

python -m pytest -q
python -m ruff check auth api ai db routers services utils main.py tests
python -m ruff format --check auth api ai db routers services utils main.py tests
cd frontend: npm test; npm run lint; npm run build

Original-paper references are listed beside each use case. Crossref API reference: https://api.crossref.org/swagger-ui/index.html . Repository: https://github.com/Aniket-Nikam/ArchimedesAI .

Read docs/OPERATIONS_REVIEW.md for backup, cancellation, credential-storage and operating boundaries. Historical automated reports describe earlier revisions and must not be substituted for this live study.